RoadMap To Robotics

STAGE 1 FOUNDATIONS
(1ST SEMESTER / ~35 DAYS)

HARDWARE

- **BASIC ELECTRONIC COMPONENTS:** RESISTORS, CAPACITORS, DIODES, TRANSISTORS
- **LOGIC ICS:** AND, OR, NOT, XOR GATES, MULTIPLEXERS, ADDERS
- **MULTIMETER** USAGE
- **SOLDERING** BASICS
- **OSCILLOSCOPE** BASICS (OPTIONAL)
- **BREADBOARD** PROTOTYPING
- **CIRCUIT CONNECTIONS** AND **PIN UNDERSTANDING**
- **READING DATASHEETS**
- **VOLTAGE REGULATORS**
- **BUCK AND BOOST CONVERTERS**
- **POWER SOURCES**: LI-ION, LIPO, AA BATTERIES
- **BASIC SENSORS**: LDR, TEMPERATURE SENSORS, PUSH BUTTONS
- BASIC ELECTRONIC **SAFETY AND POLARITY**

PROTOTYPING

- **BREADBOARD CIRCUITS**
- USE OF **PROTOTYPING MATERIALS** (SUNBOARD, ACRYLIC, CARDBOARD, STYROFOAM)

SOFTWARE

- PROGRAMMING BASICS
- **C++** FUNDAMENTALS
- **PYTHON** FUNDAMENTALS
- **OBJECT ORIENTED PROGRAMMING BASICS**
- **CONTROL STRUCTURES AND LOGIC BUILDING**
- **CODE EDITORS**: VS CODE
- **GIT** BASICS
- **GITHUB USAGE AND REPOSITORIES**
- **CLI BASICS**
- **BASIC CAD TOOLS**: TINKERCAD, FREECAD, SKETCHUP

STAGE 2 MICROCONTROLLERS & EMBEDDED BASICS
(2ND SEMESTER / ~45 DAYS)

HARDWARE

- **MICROCONTROLLERS:** ARDUINO UNO, ESP32, ESP8266
- **GPIO PROGRAMMING**
- **PWM CONTROL**
- **ANALOG VS DIGITAL SIGNALS**
- **SENSOR INTEGRATION**
- **SENSOR KITS** EXPLORATION

SENSORS

- **IMU SENSORS (MPU6050)**

ACTUATORS

- **DC MOTORS**
- **DC MOTORS WITH ENCODERS**
- **SERVO MOTORS**
- **STEPPER MOTORS**

MOTOR DRIVERS

- **L298N**
- **TB6612FNG**
- **A4988**
- **DRV8825**

POWER SYSTEMS

- **MOTOR POWER SUPPLY MANAGEMENT**
- **VOLTAGE REGULATORS**
- **BATTERY PROTECTION MODULES**

COMMUNICATION PROTOCOLS

- **UART**
- **I2C**
- **SPI**
- **WIFI**
- **BLE**
- **LORA**
- **ZIGBEE**
- **MQTT BASICS**

SOFTWARE

- **ARDUINO IDE**
- **PLATFORMIO**
- **EMBEDDED C/C++ BASICS**
- **LIBRARY USAGE AND INTEGRATION**
- **SERIAL DEBUGGING**
- **PYTHON ROBOTICS TOOLS**
- **OPENCV BASICS**
- **MEDIAPIPE BASICS**
- **TENSORFLOW LITE BASICS**
- **API USAGE BASICS**

STAGE 3 — ADVANCED EMBEDDED SYSTEMS & AI
(3RD SEMESTER / ~85 DAYS)

HARDWARE BOARDS

- STM32 BOARDS
- RASPBERRY PI PICO
- RASPBERRY PI ZERO 2W
- **RASPBERRY PI 4 / 5**
- NVIDIA JETSON NANO
- JETSON NANO (OPTIONAL)

EDGE AI HARDWARE

- CORAL TPU
- INTEL MOVIDIUS NEURAL COMPUTE STICK

ADVANCED SENSORS

- **LIDAR SENSORS**
- STEREO CAMERAS
- DEPTH CAMERAS
- **GPS MODULES**
- ADVANCED IMUS

COMPUTER VISION

- **CAMERA INTERFACING**
- IMAGE PROCESSING BASICS
- OBJECT DETECTION MODELS
- **EDGE AI INFERENCE**

SOFTWARE

- LINUX FUNDAMENTALS
- TERMINAL / CLI USAGE
- FILE SYSTEM MANAGEMENT
- SSH USAGE
- PACKAGE MANAGERS
- PYTHON FOR ROBOTICS
- OPENCV ADVANCED USAGE
- TENSORFLOW / PYTORCH BASICS
- **AI MODEL DEPLOYMENT ON EDGE DEVICES**

ELECTRONICS DESIGN

- PCB DESIGN BASICS
- EASYEDA
- KICAD

MECHANICAL DESIGN

- CAD DESIGN FUNDAMENTALS
- FUSION 360
- ONSHAPE
- MECHANICAL ASSEMBLIES
- 3D PRINTING BASICS
- TOLERANCES AND FIT

STAGE 4 — ROBOTICS SYSTEMS & INTEGRATION
(4TH SEMESTER / ~150 DAYS)

HARDWARE

- INDUSTRIAL SENSORS
- ROTARY ENCODERS
- LIDAR SYSTEMS
- DEPTH CAMERAS
- SLAM SENSORS

ACTUATORS

- CLOSED LOOP SERVO MOTORS
- BRUSHLESS MOTORS (BLDC)
- ROBOTIC ARM ACTUATORS

MECHANICAL

- FULL ROBOT MECHANICAL DESIGN
- CHASSIS DESIGN
- 3D PRINTED STRUCTURES
- ROBOT PROTOTYPING

SOFTWARE

- ROS / ROS2 BASICS
- NODES
- TOPICS
- SERVICES
- ACTIONS
- PARAMETERS

ROBOTICS ALGORITHMS

- SLAM BASICS
- LOCALIZATION
- PATH PLANNING
- MOTION CONTROL

SIMULATION

- GAZEBO
- RVIZ
- ISAAC SIM

ROBOTICS PACKAGES

- MOVEIT
- NAV2
- SENSOR INTEGRATION PACKAGES

ADVANCED COMMUNICATION

- DDS
- RTPS
- CAN BUS
- CAN-FD
- ETHERCAT

DIGITAL ROBOTICS

- DIGITAL TWINS
- SIMULATION TESTING
- HARDWARE–SOFTWARE INTEGRATION

TIPS: FOR LEARNING ROBOTICS

- ***BUILD WHILE LEARNING –*** *DON'T WAIT UNTIL YOU KNOW EVERYTHING. SMALL PROJECTS HELP CONCEPTS STICK BETTER THAN ONLY STUDYING THEORY.*
- ***START SIMPLE –*** *LED BLINKING, SENSOR READING, AND SMALL MOTOR CONTROL PROJECTS ARE ENOUGH IN THE BEGINNING.*
- ***READ DATASHEETS EARLY –*** *MANY BEGINNERS AVOID DATASHEETS, BUT LEARNING TO READ THEM WILL SOLVE MANY HARDWARE PROBLEMS.*
- ***DOCUMENT YOUR PROJECTS –*** *UPLOAD CODE, WIRING DIAGRAMS, AND PROJECT NOTES TO GITHUB.*
- ***BREAK PROBLEMS INTO SMALL PARTS –*** *ROBOTICS SYSTEMS ARE COMPLEX, BUT EACH PART (SENSOR, MOTOR, CONTROL LOGIC) CAN BE SOLVED STEP BY STEP.*
- ***DON'T DEPEND ONLY ON TUTORIALS –*** *TRY MODIFYING PROJECTS AND EXPERIMENTING ON YOUR OWN.*
- ***LEARN DEBUGGING SKILLS –*** *USE SERIAL PRINTS, MULTIMETER CHECKS, AND LOGIC TESTING TO FIND ISSUES.*
- ***REUSE EXISTING LIBRARIES –*** *MANY ROBOTICS PROBLEMS ARE ALREADY SOLVED IN OPEN-SOURCE LIBRARIES.*
- ***JOIN ROBOTICS COMMUNITIES –*** *FORUMS, GITHUB, DISCORD SERVERS, AND ROBOTICS GROUPS HELP WHEN YOU GET STUCK.*
- ***FOCUS ON FUNDAMENTALS –*** *ELECTRONICS BASICS, PROGRAMMING LOGIC, AND MATHEMATICS MATTER MORE THAN BUYING EXPENSIVE HARDWARE.*
- ***WORK ON REAL PROJECTS –*** *EXAMPLES: LINE FOLLOWER ROBOT, OBSTACLE AVOIDING ROBOT, ROBOTIC ARM, VISION-BASED ROBOT.*
- ***LEARN LINUX EARLY –*** *MANY ROBOTICS TOOLS AND FRAMEWORKS RUN ON LINUX SYSTEMS.*
- ***DON'T RUSH THE STAGES –*** *EACH STAGE BUILDS THE FOUNDATION FOR THE NEXT ONE.*
- ***EXPECT FAILURES –*** *CIRCUITS WILL FAIL, CODE WILL BREAK, AND SENSORS WILL MISBEHAVE. THAT IS NORMAL IN ROBOTICS.*
- ***STAY CONSISTENT –*** *EVEN 1–2 HOURS OF WORK EVERY DAY BUILDS STRONG SKILLS OVER TIME.*

CHOOSE ONE PATH AND START ENHANCING YOUR SKILLS

- **ROS DEVELOPER**
- **ROBOTICS AI ENGINEER**
- **EMBEDDED SYSTEMS ENGINEER**
- **ROBOTICS MECHANICAL DESIGNER**
- **COMPUTER VISION ENGINEER**
- **AUTONOMOUS SYSTEMS ENGINEER**
- **DIGITAL TWIN / SIMULATION ENGINEER**

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